



GIRRAWEE HIGH SCHOOL

YEAR 11 - TASK 4

2008

MATHEMATICS
2 UNIT

Time allowed – 90 minutes

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Start each question on a new sheet of paper.

Question 1 (10 marks)**Marks**

(a) Evaluate

$$(i) \quad \lim_{x \rightarrow 4} \frac{2x-3}{x} \quad 1$$

$$(ii) \quad \lim_{x \rightarrow -3} \frac{x+3}{x^2-9} \quad 2$$

$$(iii) \quad \lim_{x \rightarrow \infty} \frac{x^3+5x^2+3}{4x^4-x^2+x} \quad 2$$

$$(iv) \quad \lim_{x \rightarrow \infty} \frac{x^3+5x^2+3}{4x^3-x^2+x} \quad 2$$

(b) If $f(x) = x^2 - 3x + 2$, differentiate $f(x)$ from first principles 3

Question 2 (15 marks)

(a) Differentiate

$$(i) \quad y = 2x^5 - 4x^2 + 21x + 7 \quad 1$$

$$(ii) \quad y = \frac{15x^5 + 6x^3 - 12x}{3x} \quad 2$$

$$(iii) \quad y = 12x^3 - \sqrt{2}x^2 + \pi \quad 2$$

$$(iv) \quad y = \sqrt{5}x + \frac{1}{x^4} \quad 2$$

$$(v) \quad y = x^{\frac{3}{5}} - 7x^{\frac{1}{4}} \quad 3$$

$$(vi) \quad y = \sqrt[4]{x^3} \quad 2$$

(b) Differentiate $f(x) = x^3 - 7x^2 - 5x + 6$ and hence find the two points on $y = x^3 - 7x^2 - 5x + 6$ where the tangent is horizontal. 3

Question 3 (7 marks)

For the function $y = x^3 - 2x^2$

- (a) Find $\frac{dy}{dx}$ 1
- (b) Find the equations of the tangent and normal lines to the curve at $(2,0)$. Give your answer in general form. 3
- (c) Find the gradient at $(a, a^3 - 2a^2)$ 1
- (d) Find the equation of the tangent at $(a, a^3 - 2a^2)$. Give your answer in gradient / intercept form. 2

Question 4 (7 marks)

- (a) Differentiate $y = (x^2 + 1)(2x^3 - 3x)$ using product rule. 3
- (b) Differentiate $y = (7x^5 - 13x^2 + 4)^7$ 2
- (c) Differentiate $y = \frac{1}{\sqrt{5x+2}}$ 2

Question 5 (13 marks)

- (a) Differentiate $y = 8x^3(4x^2 - 9)^5$ 4
- (b) Differentiate $y = \sqrt{5-x}(10-x)^3$ 5
- (c) Find $\frac{dy}{dx}$ for $y = (2x+5)^2(x+6)^2$ and find where $\frac{dy}{dx} = 0$ 4

Question 6 (12 marks)

(a) Differentiate the following

(i) $f(x) = \frac{x}{4x-7}$ 3

(ii) $y = \frac{(2x+3)^3}{6x+5}$ 4

(b) Find the point on $y = \frac{1}{\sqrt{x+1}}$ where the tangent is
parallel to $x + 128y + 256 = 0$ 5

Question 7 (10 marks)

(a) Find the acute angle (to the nearest minute) at which the tangent
of $y = 5x^4 - 6x^2 + 4$ at $\left(\frac{1}{2}, \frac{45}{16}\right)$ will intersect the x -axis 3

(b) For $f(x) = \frac{(3x-2)^2}{4x^5}$, find

(i) $f'(x)$ 4

(ii) $f'(1)$ 1

(iii) $f'(x+2)$ 2

Q1

$$a) i) \lim_{x \rightarrow 4} \frac{2x-3}{4}$$

$$= \frac{2(4)-3}{4}$$

$$= 5/4$$

(1)

$$ii) \lim_{x \rightarrow -3} \frac{x+3}{x^2-9}$$

$$= \lim_{x \rightarrow -3} \frac{x+3}{(x+3)(x-3)}$$

$$= \lim_{x \rightarrow -3} \frac{1}{x-3}$$

$$= -1/6$$

(2)

$$iii) \lim_{x \rightarrow \infty} \frac{x^3 + 5x^2 + 3}{4x^4 - x^2 + x} \cdot \frac{1/x^4}{1/x^4}$$

$$= \lim_{x \rightarrow \infty} \frac{1/x + 5/x^2 + 3/x^4}{4 - 1/x^2 + 1/x^3}$$

$$= \frac{0 + 0 + 0}{4 - 0 + 0}$$

$$= 0$$

(2)

$$iv) \lim_{x \rightarrow \infty} \frac{x^3 + 5x^2 + 3}{4x^3 - x^2 + x} \cdot \frac{1/x^3}{1/x^3}$$

$$= \lim_{x \rightarrow \infty} \frac{1 + 5/x + 3/x^3}{4 - 1/x + 1/x^2}$$

$$= \frac{1 + 0 + 0}{4 - 0 + 0}$$

$$= 1/4$$

(2)

$$b) f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)^2 - 3(x+h) + 2 - (x^2 - 3x + 2)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2hx + h^2 - 3x - 3h + 2 - x^2 + 3x - 2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2hx + h^2 - 3h}{h}$$

$$= \lim_{h \rightarrow 0} 2x + h - 3$$

$$= 2x - 3$$

(3)

Q2 i) $y' = 10x^4 - 8x + 21$ (1)

a) ii) $y = 5x^4 + 2x^2 - 4$
 $y' = 20x^3 + 4x$ (2)

iii) $y' = 36x^2 - 2\sqrt{2}x$ (2)

iv) $y = \sqrt{5}x + x^{-4}$
 $y' = \sqrt{5} - 4x^{-5}$ (2)
 $= \sqrt{5} - \frac{4}{x^5}$

v) $y' = \frac{3}{5}x^{-2/5} - \frac{7}{4}x^{-3/4}$
 $= \frac{3}{5x^{2/5}} - \frac{7}{4x^{3/4}}$ (3)

vi) $y = x^{3/4}$
 $y' = \frac{3}{4}x^{-1/4}$
 $= \frac{3}{4\sqrt[4]{x}}$ (2)

b) $f(x) = 3x^2 - 14x - 5$
 $= (3x + 1)(x - 5)$

$0 = (3x + 1)(x - 5)$ (3)

$\therefore x = -\frac{1}{3} \quad x = 5$

$f(-\frac{1}{3}) = (-\frac{1}{3})^3 - 7(-\frac{1}{3})^2 - 5(-\frac{1}{3}) + 6$
 $= -\frac{1}{27} - \frac{7}{9} + \frac{5}{3} + 6$
 $= -\frac{1}{27} - \frac{21}{27} + \frac{45}{27} + \frac{162}{27}$
 $= \frac{185}{27} \quad \therefore (-\frac{1}{3}, \frac{185}{27})$

$f(5) = 5^3 - 7(5)^2 - 5(5) + 6$
 $= 125 - 175 - 25 + 6$
 $= -69 \quad \therefore (5, -69)$

Q3 a) $\frac{dy}{dx} = 3x^2 - 4x$ (1)

b) $\left. \frac{dy}{dx} \right|_{x=2} = 3(2)^2 - 4(2)$
 $= 12 - 8$
 $= 4$

TANGENT:

$$y - y_1 = m(x - x_1)$$

$$y = 4(x - 2)$$

$$0 = 4x - y - 8$$

(3)

NORMAL:

$$y - y_1 = m(x - x_1)$$

$$y = -\frac{1}{4}(x - 2)$$

$$x + 4y - 2 = 0$$

c) $\left. \frac{dy}{dx} \right|_{x=a} = 3a^2 - 4a$ (1)

d) TANGENT:

$$y - (a^3 - 2a^2) = (3a^2 - 4a)(x - a)$$

$$y = (3a^2 - 4a)x - 3a^3 + 4a^2 + a^3 - 2a^2$$

$$y = (3a^2 - 4a)x + (-2a^3 + 2a^2)$$

(2)

Q4

$$u = x^2 + 1 \quad v = 2x^3 - 3x$$

$$u' = 2x \quad v' = 6x^2 - 3$$

a)

$$y' = vu' + uv'$$

$$= (2x^3 - 3x)(2x) + (x^2 + 1)(6x^2 - 3)$$

$$= 4x^4 - 6x^2 + 6x^4 - 3x^2 + 6x^2 - 3$$

$$= 10x^4 - 3x^2 - 3$$

(3)

b) $y' = 7(7x^5 - 13x^4 + 4)^6 (35x^4 - 26x)$ (2)

c) $y = (5x + 2)^{1/2}$

$$y' = \frac{1}{2}(5x + 2)^{-1/2} \cdot 5$$

(2)

$$= \frac{5}{2\sqrt{5x + 2}}$$

Q5 a) $u = 8x^3$ $v = (4x^2 - 9)^5$
 $u' = 24x^2$ $v' = 5(4x^2 - 9)^4(8x)$
 $= 40x(4x^2 - 9)^4$

$$\begin{aligned} y' &= vu' + uv' \\ &= (4x^2 - 9)^5(24x^2) + 8x^3 \cdot 40x(4x^2 - 9)^4 \\ &= 8x^2(4x^2 - 9)^4 [3(4x^2 - 9) + 40x^2] \\ &= 8x^2(2x+3)^4(2x-3)^4 [12x^2 - 27 + 40x^2] \\ &= 8x^2(2x+3)^4(2x-3)^4(52x^2 - 27) \end{aligned} \quad (4)$$

b) $u = \sqrt{5-x} = (5-x)^{1/2}$ $v = (10-x)^3$
 $u' = \frac{-1}{2\sqrt{5-x}}$ $v' = -3(10-x)^2$

$$\begin{aligned} y' &= vu' + uv' \\ &= (10-x)^3 \frac{(-1)}{2\sqrt{5-x}} + \sqrt{5-x} (-3(10-x)^2) \\ &= \frac{-(10-x)^3}{2\sqrt{5-x}} + \frac{2\sqrt{5-x} \sqrt{5-x} (-3(10-x)^2)}{2\sqrt{5-x}} \\ &= \frac{(10-x)^2 [- (10-x) - 6(5-x)]}{2\sqrt{5-x}} \\ &= \frac{(10-x)^2 (7x - 40)}{2\sqrt{5-x}} \end{aligned} \quad (5)$$

c) $u = (2x+5)^2$ $v = (x+6)^2$
 $\frac{du}{dx} = 4(2x+5)$ $\frac{dv}{dx} = 2(x+6)$
 $\frac{dy}{dx} = v \frac{du}{dx} + u \frac{dv}{dx}$
 $= (x+6)^2 \cdot 4(2x+5) + (2x+5)^2 \cdot 2(x+6)$
 $= 2(x+6)(2x+5) [2(x+6) + (2x+5)]$
 $= 2(x+6)(2x+5)(4x+17)$

$\therefore \frac{dy}{dx} = 0$ WHEN $x = -6$ $y = 0$
 $x = -\frac{5}{2}$ $y = 0$
 $x = -\frac{17}{4}$ $y = \frac{2401}{64}$

Q6 i) $u = x$ $v = 4x-7$
 $u' = 1$ $v' = 4$

a) $f'(x) = \frac{v u' - u v'}{v^2}$
 $= \frac{4x-7 - 4x}{(4x-7)^2}$ (3)
 $= \frac{-7}{(4x-7)^2}$

ii) $u = (2x+3)^3$ $v = 6x+5$
 $u' = 3(2x+3)^2 \cdot 2$ $v' = 6$
 $= 6(2x+3)^2$

$y' = \frac{v u' - u v'}{v^2}$
 $= \frac{(6x+5) 6(2x+3)^2 - (2x+3)^3 \cdot 6}{(6x+5)^2}$
 $= \frac{6(2x+3)^2 [6x+5 - (2x+3)]}{(6x+5)^2}$
 $= \frac{6(2x+3)^2 (4x+2)}{(6x+5)^2}$ (4)
 $= \frac{12(2x+3)^2 (2x+1)}{(6x+5)^2}$

b) $y = (x+1)^{-1/2}$
 $y' = \frac{-1}{2} (x+1)^{-3/2}$
 $= \frac{-1}{2\sqrt{x+1}^3}$

$x + 128y + 256 = 0$

$128y = -x - 256$

$y = \frac{-1}{128}x - 2$

$\therefore m = \frac{-1}{128}$

$\therefore \frac{-1}{128} = \frac{-1}{2\sqrt{x+1}^3}$

$-128 = -2\sqrt{x+1}^3$

$64 = (x+1)^{3/2}$

$64^{2/3} = x+1$ (5)

$16 - 1 = x$

$15 = x$

$\therefore y = \frac{1}{\sqrt{15+1}} \quad \therefore (15, \frac{1}{4})$
 $= \frac{1}{4}$

Q7 $y' = 20x^3 - 12x$

a) $y' \Big|_{x=\frac{1}{2}} = 20\left(\frac{1}{2}\right)^3 - 12\left(\frac{1}{2}\right)$

$$= \frac{20}{8} - \frac{12}{2}$$

$$= -\frac{7}{2}$$

(3)

$$\tan^{-1}\left(\frac{7}{2}\right) = 74^\circ 3'$$

b) $u = (3x-2)^2 \quad v = 4x^5$
 $u' = 2(3x-2) \cdot 3 \quad v' = 20x^4$
 $= 6(3x-2)$

$$f'(x) = \frac{vu' - uv'}{v^2}$$

$$= \frac{4x^5(6(3x-2)) - (3x-2)^2 20x^4}{(4x^5)^2}$$

$$= \frac{4x^4(3x-2)[6x - 5(3x-2)]}{16x^{10}}$$

$$= \frac{(3x-2)(-9x+10)}{4x^6}$$

$$= \frac{(3x-2)(10-9x)}{4x^6}$$

$$f'(1) = \frac{(3(1)-2)(10-9(1))}{4(1)^6}$$

$$= \frac{1}{4}$$

$$f'(x+2) = \frac{(3(x+2)-2)(10-9(x+2))}{4(x+2)^6}$$

$$= \frac{(3x+4)(-9x-8)}{4(x+2)^6}$$

$$= -\frac{(3x+4)(9x+8)}{4(x+2)^6}$$